Energy Shocks and the Green Adoption Channel according to E-HANK

Model, experiments, and caveats

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Abstract

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of Auclert et al. (2024). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks, and the two fiscal responses studied in the literature: a retail price cap (Langot et al., 2026) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption (Bayer et al., 2026). Our central result is an interaction neither paper can see: **the price cap stabilises output but mechanically eliminates the green adoption margin**, while the transfer delivers comparable stabilisation and leaves adoption fully intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. The loss is linear in the cap's intensity under a price shock and convex under a supply shock, and the *sign* of the cap's welfare effect flips with the elasticity of energy supply — reconciling Langot et al. (2026) and Bayer et al. (2026) within a single model. The *ordering* of policies is robust; the *magnitude* of the adoption channel is not yet pinned down, because the logit taste scale governing the adoption elasticity is not identified by the moments used so far, and because the channel is strongly non-linear at the shock sizes used in this literature. Section 5 quantifies both problems and Section 3.1 proposes the moment that closes the first.

1 Contribution

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. Auclert et al. (2024) study a small open economy and stress the negative externality when all importers subsidise simultaneously. Bayer et al. (2026) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. Langot et al. (2026) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well.

All three treat the household's energy technology as fixed. We relax exactly that. The green durable is an extensive margin: the energy shock raises the return to adoption, so the crisis itself accelerates the transition — unless policy removes the price signal that drives it.

Our energy-market closure nests the disagreement between Bayer et al. (2026) and Langot et al. (2026) in a single parameter: with $\varepsilon^E = \infty$ the economy is a price taker (elastic supply, Langot); with ε^E finite the quantity is fixed and the world price clears the market (Bayer). Both are reported below.

2 Model

2.1 Environment

The aggregate structure is Auclert et al. (2024): a small open economy with sticky wages, a CES consumption basket nesting energy against a home/foreign bundle, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share ζ^E of the energy endowment.

Unit of account. The household block is written generically in the unit of account. It reads a price of the unit of account relative to the CPI, p^{num}, a real return $1+r^{num} = (1+r)\, p^{num}_{-1}/p^{num}$, and after-tax labour income $atw^{num} = atw/p^{num}$; it returns asset holdings denominated in the same unit, converted by $A^{CPI} = A\, p^{num}$ before meeting the CPI-denominated objects in asset market clearing and the current account. Two choices are implemented:

$$p^{num} = 1 \;\; \text{(ARS's CPI basket)}, \qquad p^{num} = p_H \;\; \text{(the domestic good)}.$$

We adopt the domestic good. The two are exact relabelings of the same economy and we verify this numerically in Section 5. Anchoring on the domestic good matters here because it makes the resource constraint independent of how the CPI weights brown and green energy, which is the measurement issue discussed next. Everything contractually written in CPI units — the fiscal transfers, the ARS subsistence term, and the switching cost ψ_g — is divided by p^{num} on the way into the budget. Keeping ψ_g CPI-denominated is a modelling choice, not a normalisation: denominating it in domestic-good units instead moves D^G_{ss} by about 9%.

2.2 The durable state

Each household carries a durable state d, the *pair* (holding entering the period, holding chosen last period), needed to charge the switching cost:

$$d \in \{BB,\ BG,\ GB,\ GG\},$$

where GB (green now, brown before) is the state that pays ψ_g. Brown is absorbing; green breaks down to brown at rate δ_g. The state BG carries zero mass by construction, so the effective state is three-point.

Stages, in forward order, are `[dep, prod, durables, consav]`: breakdown, idiosyncratic productivity, the discrete adoption choice (a logit with taste scale σ_ε), and the EGM consumption–savings step. The adoption stage is the discrete-continuous structure of Iskhakov et al. (2017).

2.3 Two explicit energy prices

We distinguish three prices, all relative to the CPI:

p^E	wholesale/retail market price of brown energy, before any subsidy
$P^E_B = (1-\tau^E)\, p^E + \tau^E \bar{p}^E$	price paid by a *brown* household
$P^E_G = \varrho\, \bar{p}^E \left(P^E_B/\bar{P}^E_B\right)^{\rho_g}$	price paid by a *green* adopter

with ϱ the steady-state green/brown ratio and ρ_g the pass-through of the fossil shock to the green price. The earlier code carried a single symbol p^E_{hh} doing double duty as (i) the brown household's energy price and (ii) the price entering the CPI; the two are now separate objects.

2.4 How the durable enters the budget

This is the substantive modelling decision. Households of different durable types face different energy prices, so they face different *consumption bundle prices*. We give each type its own exact CES cost-of-living index:

$$p^E_d = P^E_B + \mathbb{I}[d \text{ green}]\,(P^E_G - P^E_B), \qquad p_d = \left[1 + \alpha_E\left((p^E_d)^{1-\eta_E} - (P^E_B)^{1-\eta_E}\right)\right]^{\frac{1}{1-\eta_E}},$$

and the budget constraint becomes $(p_d/p^{num})\, c + a' = \mathrm{coh}(d)$, with coh carrying only the switching cost. The second expression uses the CES price identity enforced by the equilibrium block, so brown households have $p_d \equiv 1$ *exactly*, at every date and with structurally zero derivatives. This matters numerically: writing p_d in the naive form leaves the identity to be recovered by composing Jacobians along the DAG, which left $\sim 7 \times 10^{-6}$ of residual and broke the nesting test.

Price bases. Every CES demand ratio must have both legs in the same base. The household's demand system uses the CPI-relative p_d, because p_d^E and the home/foreign index are CPI-relative; only the budget constraint uses p_d/p^{num}. Mixing the two rescales every demand by p^{num} and breaks asset market clearing — an error that is invisible at the steady state, where $p_H = 1$.

What this replaces. An earlier version booked the energy price gap as a first-order income transfer with energy quantity frozen at its steady state. That is valid to first order but shuts down the intensive margin by construction: adopters could not re-optimise energy use. The CES formulation above is exact and lets adopters respond to their own price.

Aggregation. With type-specific prices, the aggregate energy demand is $\sum_d c_E(d)$, *not* the CES demand evaluated at the aggregate index — these differ by Jensen's inequality (0.5% here). Aggregate c_H, c_F, c_E are therefore built from the household block's own hetoutputs. Getting this wrong breaks asset market clearing by 0.002–0.06.

2.5 What the CPI measures

With two energy prices in the economy, the energy leg of the CPI could be anchored on the brown price alone or on the share-weighted index $\big[(1 - D^G)(P_B^E)^{1-\eta_E} + D^G(P_G^E)^{1-\eta_E}\big]$. **We anchor on the brown price**, for three reasons. First, it is what statistical agencies do: energy baskets are not reweighted quarterly for technology adoption. Second, it preserves $p_d \equiv 1$ for brown households exactly, which is what makes the channel-shut model reproduce Auclert et al. (2024) to machine precision. Third, feeding D^G back into the CPI closes a DAG cycle (household $\to D^G \to$ CPI $\to P_B^E \to$ household) that would require promoting the CPI to a model unknown.

Because the unit of account is the domestic good, this choice does *not* touch the resource constraint or any real allocation. It affects only the measured deflator and, through it, the monetary rule. The measurement gap has a closed form: using the CES identity enforced at every date, the ratio of the share-weighted CPI to the model's CPI is

$$\phi^{1-\eta_E} = 1 + \alpha_E\, D^G \Big[(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}\Big],$$

which involves neither the home/foreign index nor any additional state. We compute ϕ ex post, outside the DAG, and report the implied difference between true and measured inflation. **This gap is not negligible**: on the price shock without policy it reaches 0.14 annualised percentage points, about 1.5% of the amplitude of measured CPI inflation (0.23 pp, 2.3%, on the supply shock). Both are smaller than at the old calibration (0.62 pp, 6.5%), because D^G moves less at $\sigma_\varepsilon = 0.05$ and ϕ above is increasing in D^G. It should be reported as a robustness exercise rather than a footnote, and it is a genuine reason to prefer the domestic-good numeraire: had the CPI been the unit of account, this measurement choice would have moved the resource constraint.

2.6 Balance of payments

Two durable-margin terms enter the current account.

- **Switching cost as an import**, $\text{imports}_{dur} = \psi_g D^{sw}$. Booking it as domestic demand fails here: with $\mu_{ss} > 1$ and capitalised profits, extra demand for the home good creates a marginal dividend owned by nobody and breaks asset clearing by its capitalised value.
- **Energy gap as a real import saving**, $(P_B^E - P_G^E)\, C_E^G$. *This is a modelling choice, not an accounting patch*: it asserts that adopters' cheaper energy is a genuine resource saving for the country, not a domestic transfer. Without it the windfall has no external counterpart and asset clearing fails by 0.5–0.7.

2.7 Policy instruments

Both instruments are the extreme (full-compensation) versions, as in Bayer et al. (2026).

$$\text{Price cap:} \quad P_{B,t}^E = (1-\tau^E) p_t^E + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow \text{retail price fixed at pre-crisis level} \tag{1}$$

$$\text{Slutsky transfer:} \quad tr_{it} = \iota\,(p_t^E - \bar{p}^E)\,\bar{c}_{E,i}, \qquad \iota = 1 \Rightarrow \text{full compensation on } \textit{pre-crisis} \text{ use} \tag{2}$$

Both are debt financed and repaid through the distortionary labour tax. The transfer base $\bar{c}_{E,i}$ is each household's steady-state energy demand; we verified it aggregates to the government's cost base to machine precision.

Why the two differ for adoption. The cap operates on P_B^E, which is exactly the price entering the brown–green gap. It therefore compresses the adoption incentive directly. The transfer is lump-sum with respect to current choices and leaves the relative price intact.

3 Calibration

Parameter	Value	Source / target
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
r, μ_{ss}	0.01, 1.03	Auclert et al. (2024)
β heterogeneity (3 types)	spread 0.06	MPC ≈ 0.30
δ_g green breakdown rate	0.05	assumption (5 yr life)
$\varrho = P_G^E / P_B^E$ steady-state gap	0.80	assumption
ψ_g switching cost	0.538	targets $D_{ss}^G = 0.05$
σ_ε taste scale	0.05	**not identified** (§3.1)
ρ_g pass-through to green price	0	**extreme case** (§6)

Table 1: Calibration. The steady state is identical across all experiments and across both units of account.

3.1 Identifying ψ_g and σ_ε

The problem is standard, not exotic. The logit probability depends on the value gap only through the ratio $\Delta_i/\sigma_\varepsilon$, and to a first approximation

$$\Delta_i \;=\; \underbrace{\mathrm{PV}_i(\text{energy saving})}_{\text{gain}} \;-\; \underbrace{u'(c_i)\,\psi_g}_{\text{cost}},$$

so $(\psi_g, \sigma_\varepsilon)$ enter the choice probabilities only through the two combinations $\psi_g/\sigma_\varepsilon$ and $1/\sigma_\varepsilon$. This is the familiar statement that in a logit demand system only the ratio of the price coefficient to the scale is identified (Berry, 1994); one moment cannot pin two parameters. Two moments are needed and two suffice. D^{sw} cannot be the second: it satisfies $D^{sw} = \delta_g D^G$ mechanically.

Moment 1 (level). $D^G_{ss} = 0.05$, as now. This pins the position of the adoption threshold in the wealth/income distribution.

Moment 2 (slope). The semi-elasticity of the adoption flow to the purchase price of the green durable — equivalently, to a purchase subsidy:

$$\varepsilon_\psi \;\equiv\; \frac{d\log D^{sw}}{d\log \psi_g} \;=\; -\,\frac{\psi_g}{\sigma_\varepsilon}\,\frac{\mathbb{E}_B[u'(c_i)\,h_i(1-h_i)]}{\mathbb{E}_B[h_i]}, \tag{3}$$

where h_i is the household's switching hazard and $\mathbb{E}_B$ is the expectation over brown households under the stationary distribution. Every object in (3) is available from a solved steady state; no additional solve is required. `calibration_moments.py` evaluates it.

Why this is the binding problem. The calibration was moved from $\psi_g = 0.253$, $\sigma_\varepsilon = 10^{-2}$ to $\psi_g = 0.538$, $\sigma_\varepsilon = 0.05$ (still holding $D^G_{ss} = 0.05$), precisely because the earlier scale implied an implausibly sharp logit (Section 5.3). **ε_ψ has not yet been recomputed at the new calibration**: `calibration_moments.py`, which evaluates (3), is not part of the current codebase snapshot and needs to be rewritten before the moment can be checked against the empirical counterparts below. Until then $\sigma_\varepsilon = 0.05$ should be read as a five-fold correction in the right direction, not as an identified value.

Candidate empirical counterparts. The natural analogues are vehicles and household heating equipment, where purchase subsidies and fuel prices provide the variation.

- *Purchase-price / subsidy variation* (direct counterpart of (3)): Beresteanu and Li (2011) for hybrid vehicles and Muehlegger and Rapson (2022) for electric vehicles.
- *Operating-cost variation* (the fuel-price version of the same moment, which identifies the same ratio through the energy saving rather than the purchase price): Busse et al. (2013) for the US and Grigolon et al. (2018) for Europe.

The point estimates must be read off the sources and converted to this model's units; we have not done that here and no number from these papers is quoted above. The conversion is not mechanical — published elasticities are typically shares of a *stock* of new purchases, whereas D^{sw} is a quarterly *flow* out of the brown population — and doing it carefully is the natural next task.

An independent discipline on ψ_g. The calibration implies a simple payback period. A green adopter saves $(1-p_G)\,c \approx 0.0081\,c$ of CPI-denominated consumption per quarter — this ratio is set by ϱ and η_E alone and is essentially unchanged by the move to $\sigma_\varepsilon = 0.05$ — so the switching cost now pays back in $\psi_g/(4 \times 0.0081\,c) \approx 16.6/c$ years, against an expected green lifetime of $1/\delta_g = 5$ years. Equating the two, the marginal adopter is a household consuming about **3.3 times the mean**, up from 1.6 at the old calibration. The direction — energy-efficient durables adopted by the better off at implicit discount rates far above market rates (Hausman, 1979; Allcott and Greenstone, 2012) — still holds, but a 3.3×-mean marginal household is a much more extreme claim about who adopts, and is itself a symptom of ψ_g having to more than double to keep $D^G_{ss} = 0.05$ once the logit is smoothed. This tension — a higher σ_ε requiring a higher ψ_g that in turn pushes the payback discipline further from plausible — should be reported explicitly once ε_ψ is identified (previous paragraph).

4 Experiments and results

Shocks. The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life (ARS). The *supply* shock is a **5%** cut in available energy for 6 quarters; the world price then clears the market endogenously. Earlier drafts used a 20% cut following Bayer et al. (2026); we reduced it because the first-order solution is not reliable at that size (Section 5).

Units. Impulse responses are reported as 100× the *level* deviation from steady state. For variables normalised to one at the steady state (y, C, n, w, p^E, $P^{E\star}$) this coincides with a percent deviation. For population shares (D^G, D^{sw}) it is percentage points: D^G moving from 0.050 to 0.258 is +20.8 pp, not "+20.8 log points". For energy quantities (c_E, E^{supply}, steady state ≈ 0.040) it is neither: divide by 0.0401 to express them as a percent of their own steady state. This convention was previously mislabelled on the figure axes and in the tables.

Reading the supply shock in the IRFs. The instrument `E_supply_shock` is a quantity calibrated to $c_E + \mathrm{prod}_E \approx 0.0402$, so a 5% cut is −0.00201 in levels. The realised path of E^{supply} is smaller than the imposed cut at impact, because the world energy stock is arbitraged intertemporally: inventories are drawn down when the price is expected to fall back, spreading the shock over more quarters.

Shock	Policy	$y(0)$%	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption → $\sum y$
price	none	−1.123	−30.06	9.14	**9.18**	0.00	−1.60
price	cap	−0.004	−20.93	−1.53	**0.01**	53.82	+0.36
price	transfer	+1.202	+12.65	11.08	**9.32**	53.39	−1.74
supply	none	−0.709	−8.16	10.06	**2.33**	0.00	+0.38
supply	cap	−0.744	−46.85	−0.96	**0.01**	26.09	+0.18
supply	transfer	+1.331	+2.72	14.46	**3.39**	26.56	+2.48

Table 2: Summary. Cumulative sums over 24 quarters, at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$. The last column is the contribution of the adoption margin, computed as the difference between the model with the margin open and the same model with it shut. The supply shock is the 5% cut described above (not rescaled from a larger figure).

4.1 Result 1: the crisis accelerates the transition — but the output effect of adoption now depends on the shock

With no policy, the green share rises by up to 9.2 percentage points (price shock, i.e. from 5.0% to 14.2% of households) and 2.3 pp (5% supply shock) — smaller swings than at the old $\sigma_\varepsilon = 10^{-2}$, as expected from smoothing the logit, but the direction is unchanged.

What did change qualitatively: at this calibration the adoption margin is no longer unambiguously expansionary. Under the **supply** shock it still is (adoption adds +0.38 points of cumulative output with no policy, +2.48 under the transfer), for the reason given before — adopters escape the energy bill and their real income falls less. Under the **price** shock the sign flips: adoption now *subtracts* 1.60 points of cumulative output with no policy and 1.74 under the transfer. We verified this directly (not a decomposition artifact): impact output is worse with the channel open than shut (−1.123% vs −0.929%), and the gap widens over the horizon. The mechanical reason is that ψ_g more than doubled (to hold $D^G_{ss} = 0.05$ against the smoother logit) and the switching cost is booked as an import in the same periods household income is falling hardest (Section 2.6); at $\psi_g = 0.538$ that resource cost now outweighs the energy-bill saving on the price shock, whereas the supply shock's smaller peak adoption flow keeps the balance the other way. **This means the "adoption is expansionary" framing in the abstract-level pitch of**

this paper is not a general property of the model — it depends on the calibration of ψ_g and on which shock is driving adoption, and should be presented as such until ε_ψ disciplines ψ_g independently (Section 3.1).

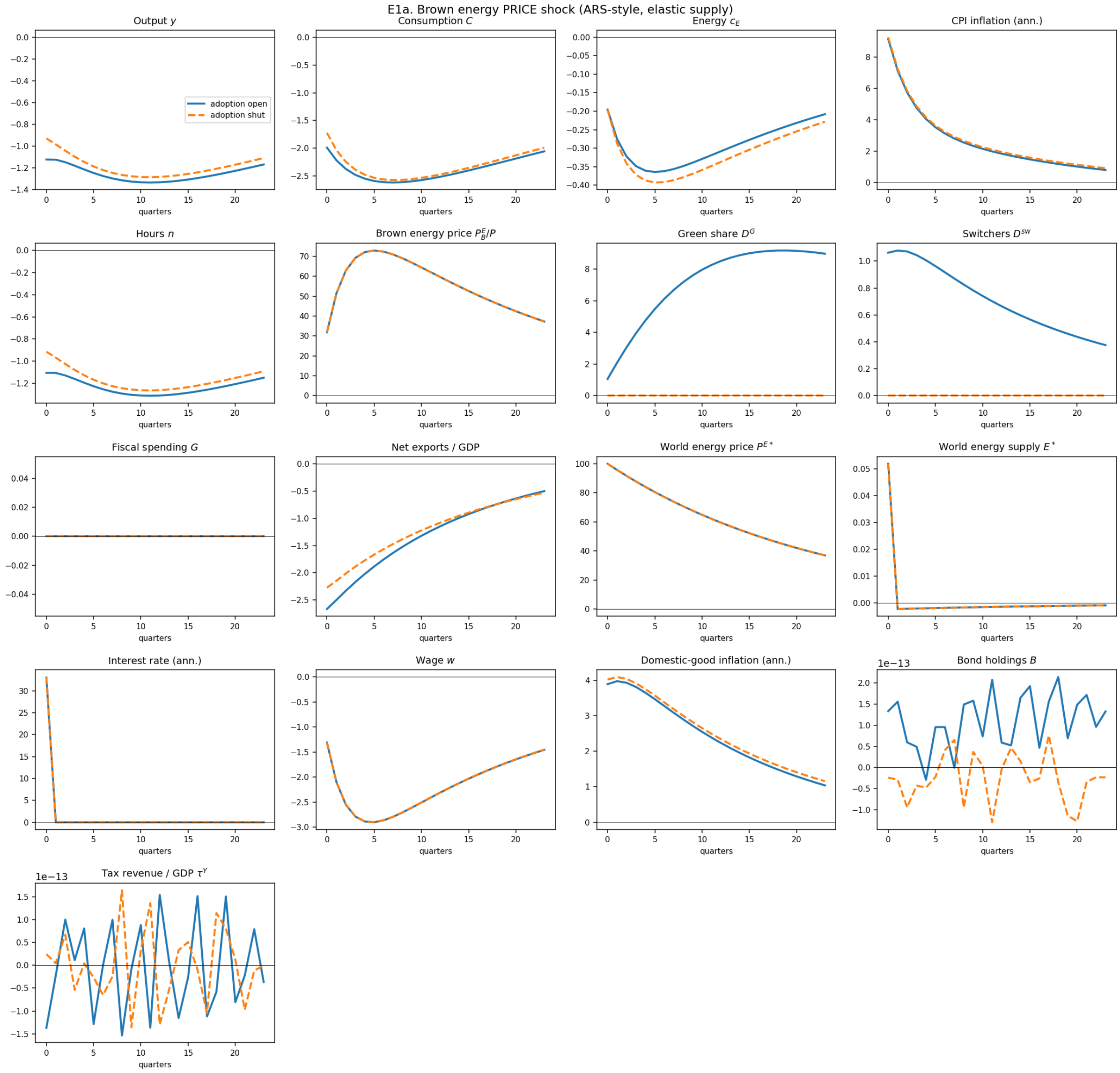

Figure 1: Brown energy PRICE shock, adoption open (solid) vs shut (dashed). Elastic energy supply, no policy.

4.2 Result 2: the price cap eliminates the adoption margin

This is the paper's headline, and it survives the recalibration. Under the cap, the peak green share response falls from +9.18 to +0.01 pp under the price shock, and from +2.33 to +0.01 under the supply shock. **The adoption channel is not attenuated; it is switched off**. Its contribution to cumulative output moves from −1.60 to +0.36 under the price shock (from +0.38 to +0.18 under the supply shock) — consistent with switching the channel off removing whichever effect, expansionary or contractionary, adoption was having. The mechanism is unchanged: the cap holds P_B^E at its pre-crisis level, and P_B^E is precisely what determines the return to adoption.

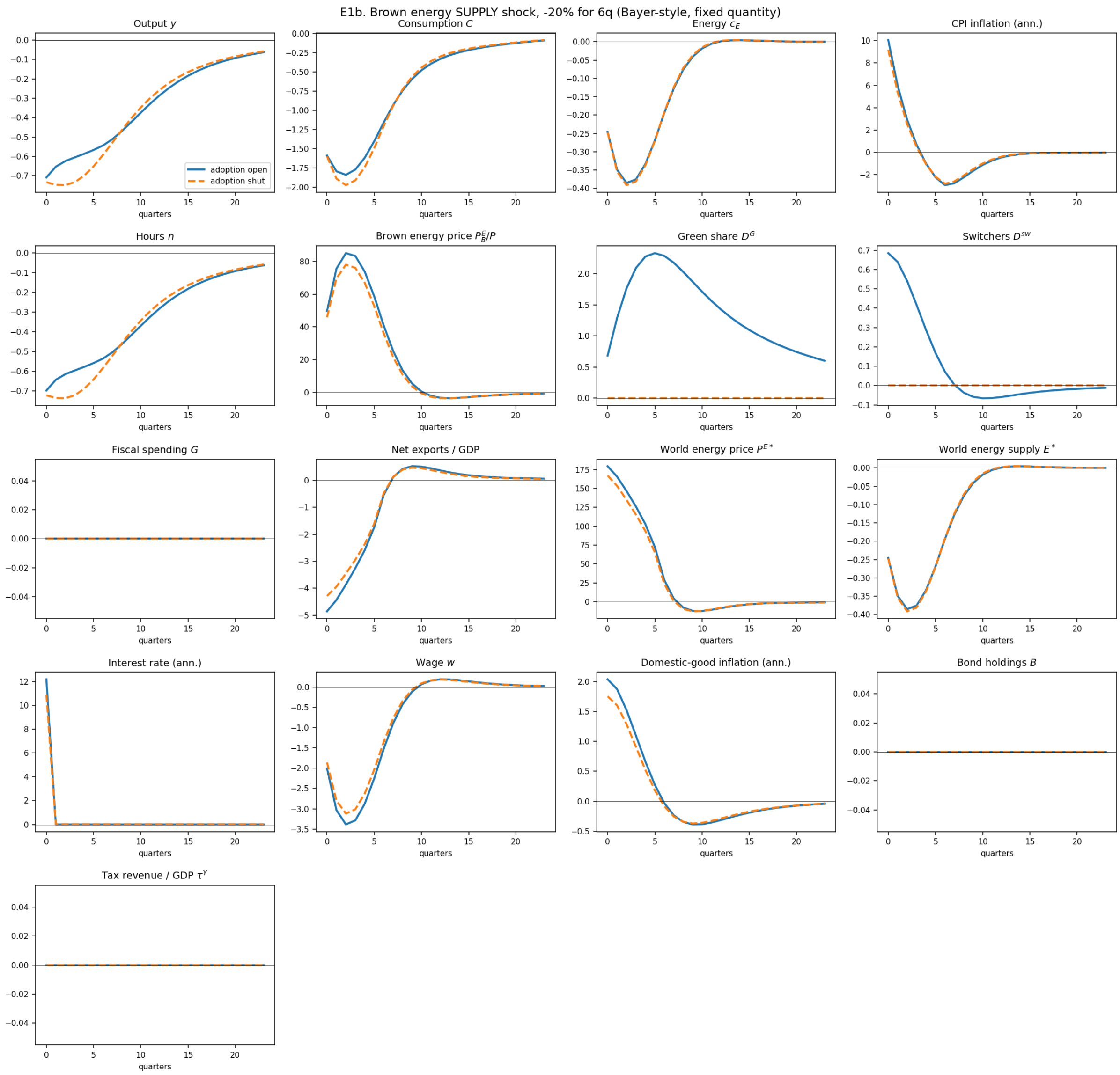

Figure 2: Brown energy SUPPLY shock, -5% for 6 quarters, adoption open (solid) vs shut (dashed). Fixed quantity, world price clears the market, no policy.

4.3 Result 3: the transfer stabilises without eliminating the margin — but no longer "without this cost"

The Slutsky transfer leaves the green share response essentially unchanged (9.32 vs 9.18 pp under the price shock, 3.39 vs 2.33 under the supply shock) while delivering more output stabilisation than the cap, at similar fiscal cost. This still strengthens Bayer et al. (2026)'s case for transfers over subsidies with an argument they do not make: subsidies also destroy the transition margin. What no longer follows automatically is that leaving adoption intact is a free lunch for output: under the price shock the transfer's cumulative-output gain is now *smaller* with adoption open than shut (-1.74 points from the margin, Table 2), for the same ψ_g-cost reason as Result 1.

4.4 Result 4: with fixed supply, the cap is destructive

Under the supply shock the cap yields cumulative output of -46.9 against -8.2 with no policy — a substantially worse gap than before. This replicates Bayer et al. (2026)'s core finding —

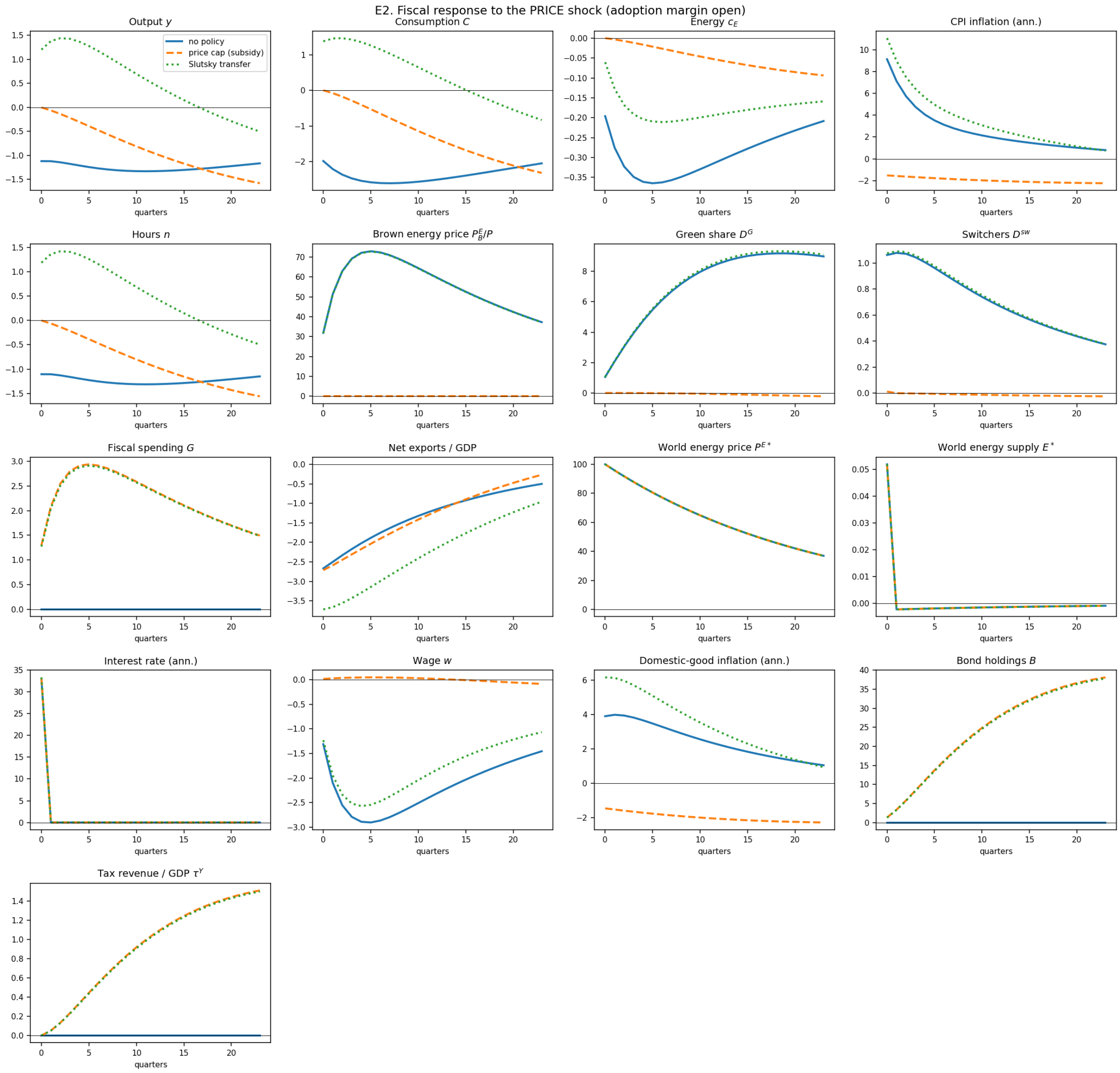

Figure 3: Fiscal response to the PRICE shock, adoption margin open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief — and shows it survives the addition of an adoption margin. It is also precisely the configuration in which Langot et al. (2026), assuming elastic supply, find the cap effective. Our elastic-supply column agrees with them (-20.9 vs -30.1 for cap vs. no policy under the price shock), and the improvement from capping is now much larger than at the old calibration ($\sim$10 points of cumulative output instead of $\sim$3).

4.5 Result 4bis: monetary policy

Experiment E4 compares the constant real-rate rule (ARS baseline) against a standard Taylor rule under the supply shock, adoption open. **Placeholder — discussion to be written once the deflator-anchoring caveat (Section 2.5) has been factored in**: under a Taylor rule the 0.23 pp measurement gap between the brown-anchored CPI and the true share-weighted deflator (supply shock, no policy; Section 2.5) feeds directly into the policy rate, so this experiment should be read jointly with Figure 9.

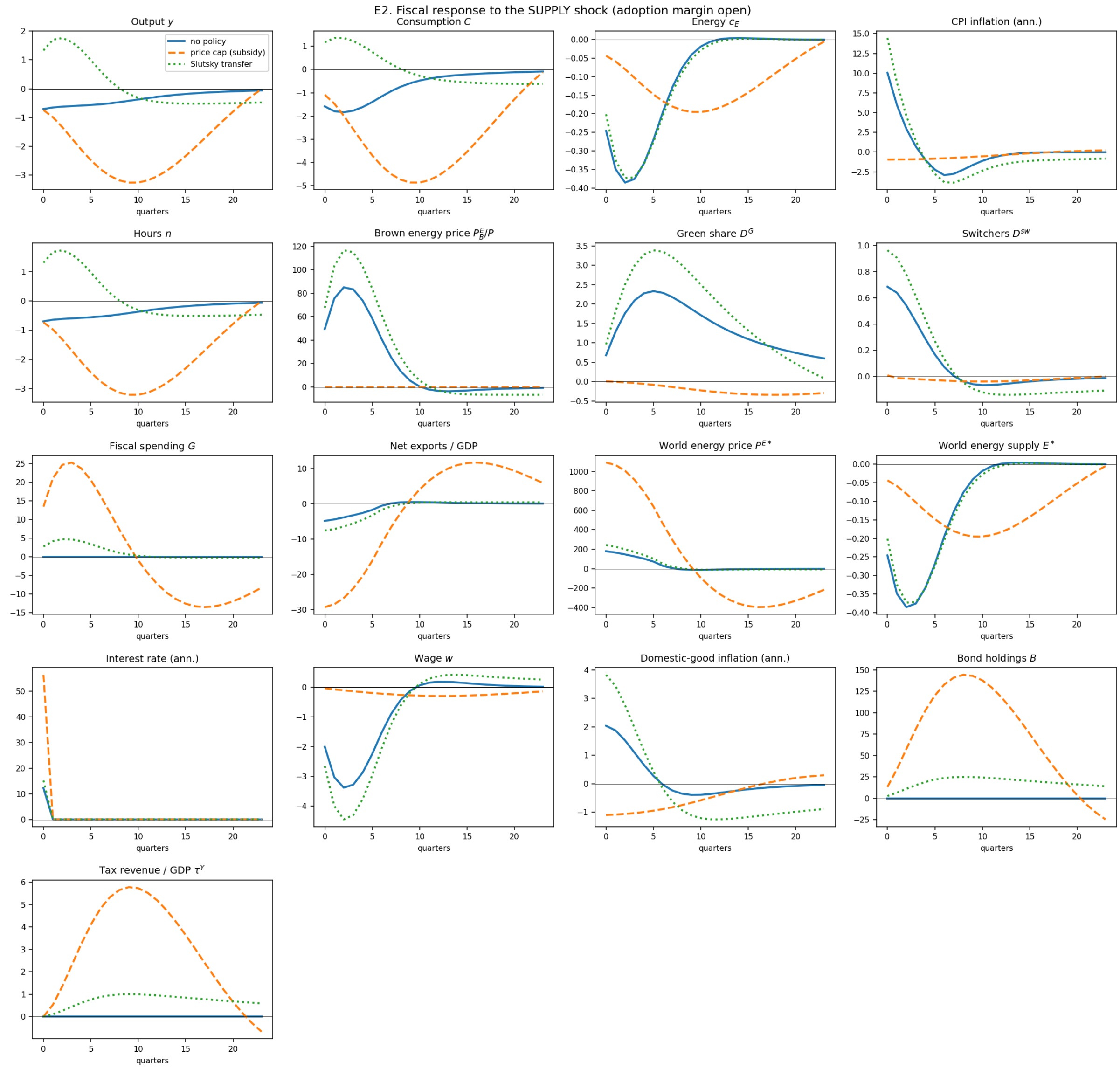

Figure 4: Fiscal response to the SUPPLY shock, adoption margin open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

4.6 Result 5: the cap is a dial, not a switch

Section 4.2 used $\tau^E = 1$, a complete price freeze. No government did that. Table 3 sweeps $\tau^E \in [0, 1]$.

Two different shapes, unchanged from the earlier calibration. Under the *price* shock the adoption loss is almost exactly **linear** in τ^E: half a cap costs a bit under half the transition (4.53 against 9.18). Under the *supply* shock it is **convex**: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial French-style shield is therefore much less damaging to the transition when energy supply is inelastic — but, as the next result shows, much more damaging to everything else. Note that this contrast is a property of the shock transmission, not of the shock size: rescaling the supply cut leaves the shape unchanged.

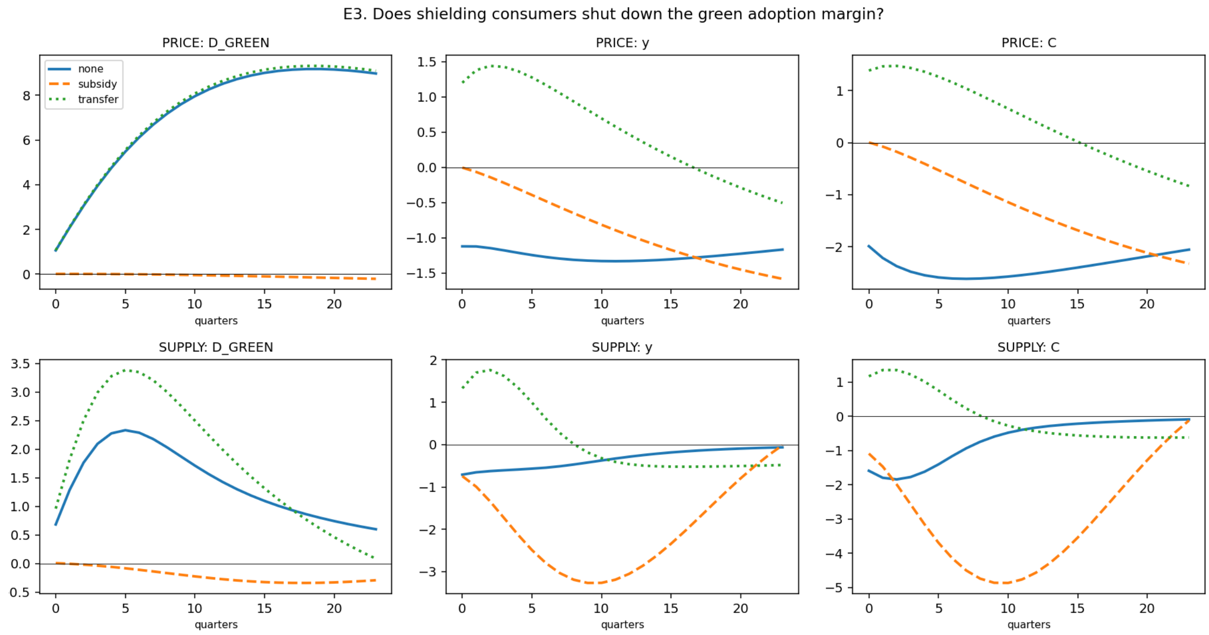

Figure 5: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

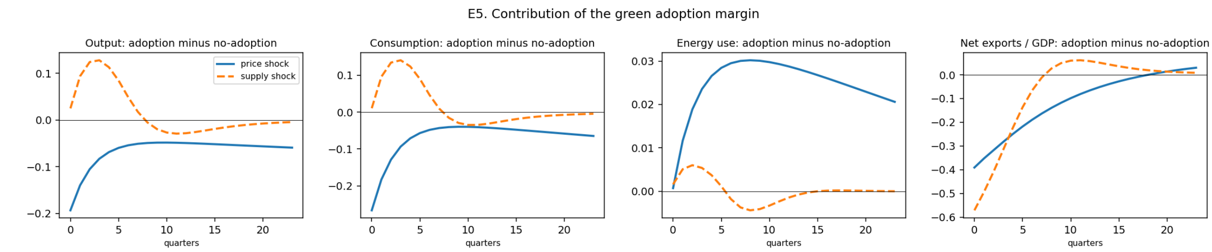

Figure 6: Contribution of the adoption margin: difference between the model with the margin open and with it shut.

4.7 Result 6: the sign of the welfare effect flips with the supply elasticity

This is the sharpest result in the paper, and it is the one result that is robust in both sign and (roughly) magnitude across the recalibration. Under the price shock, tightening the cap *raises* welfare monotonically, from −1.30% to −0.57%. Under the supply shock it *lowers* welfare monotonically, from −0.59% to −1.28% — more than doubling the cost of the crisis.

Langot et al. (2026) assume elastic supply and conclude the French shield worked. Bayer et al. (2026) assume fixed supply and conclude subsidies are destructive. **Both are right within their own closure, and our model reproduces each by moving a single parameter.** The disagreement in the literature is not about mechanisms; it is an empirical question about the elasticity of energy supply at the relevant horizon.

4.8 Result 7: distributional incidence — the reversal does not survive the recalibration

Without policy the impatient still lose far more than the patient (−2.10 vs −0.14, roughly 15×, an even sharper gap than before). **At this calibration the full cap does not reverse the incidence**: the patient remain the least-hurt group throughout the sweep (−0.08 at $\tau^E = 1$, against −0.78 for the impatient). What does happen is that the *impatient* and *middle* types cross

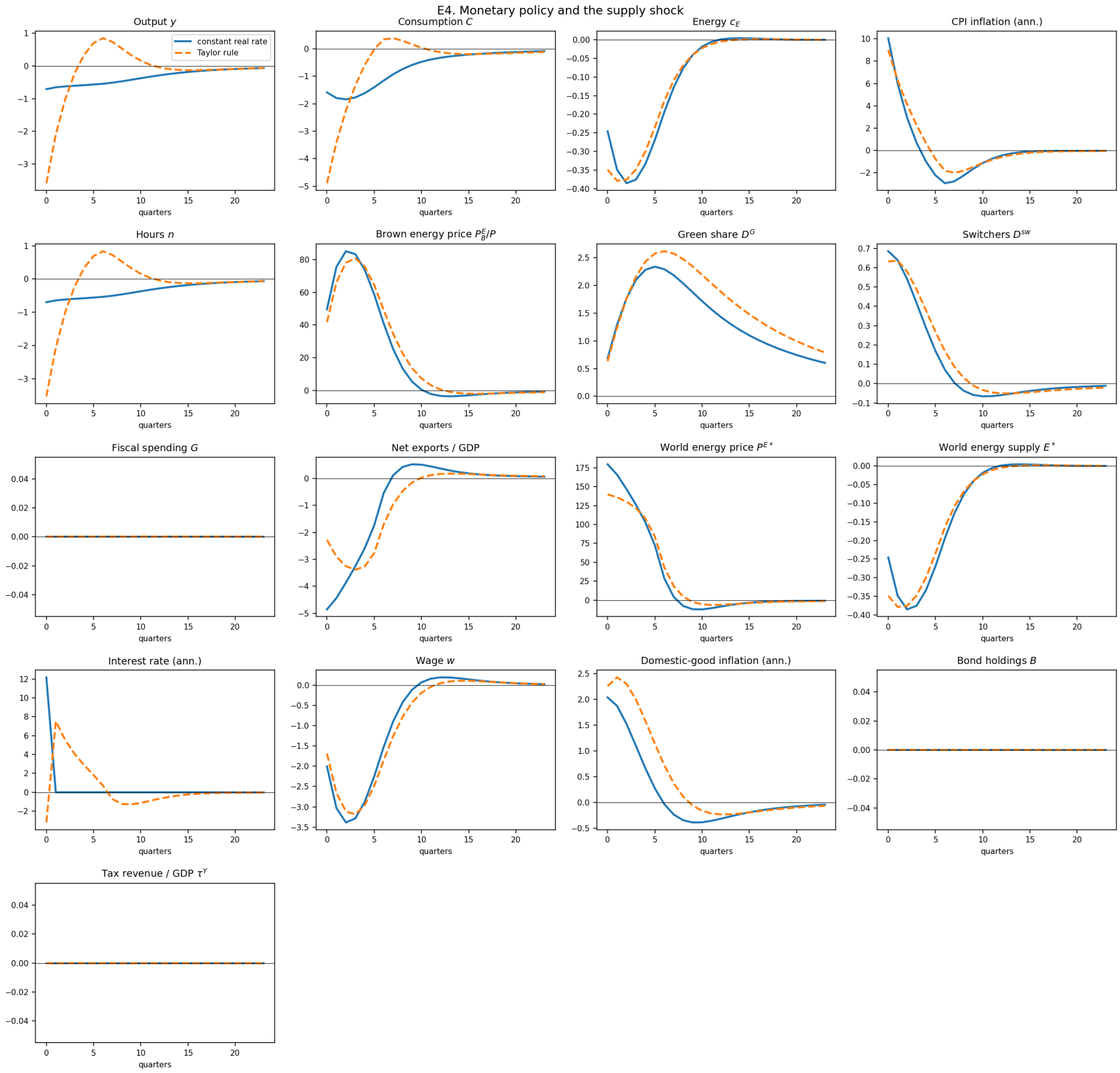

Figure 7: Monetary policy and the supply shock: constant real rate (solid) vs Taylor rule (dashed), adoption margin open.

— the impatient go from losing the most (tied with middle) at $\tau^E = 0$ to losing slightly *less* than the middle type by $\tau^E = 1$ — but this is a much narrower and less quotable result than a full incidence reversal against the patient, and it no longer reproduces Bayer et al. (2026)'s finding the way the previous draft claimed. One clean result does survive: under the transfer, the **patient type's CEV turns positive** (+0.17%) while impatient and middle remain negative — the Slutsky compensation, indexed to pre-crisis energy use, over-compensates low-MPC households relative to their true loss. This is a sharper, more credible version of the same mechanism the previous draft was gesturing at, and is worth building the distributional discussion around instead of the now-absent reversal.

5 Solution accuracy and the limits of linearisation

This section reports numerical work that a referee will ask for and that materially qualifies the magnitudes above.

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	fiscal	CEV %
price	cap $\tau^E = 0$	9.18	-30.1	9.14	0.00	-1.300
price	cap $\tau^E = 0.25$	6.86	-27.8	6.50	13.39	-1.120
price	cap $\tau^E = 0.50$	4.53	-25.5	3.84	26.83	-0.938
price	cap $\tau^E = 0.75$	2.20	-23.2	1.17	40.30	-0.755
price	cap $\tau^E = 1$	0.01	-20.9	-1.53	53.82	-0.571
price	transfer	9.32	$+12.7$	11.08	53.39	-0.213
supply	cap $\tau^E = 0$	2.33	-8.16	10.06	0.00	-0.586
supply	cap $\tau^E = 0.25$	2.21	-10.60	9.28	5.96	-0.627
supply	cap $\tau^E = 0.50$	1.99	-14.80	8.07	15.34	-0.697
supply	cap $\tau^E = 0.75$	1.50	-23.27	5.89	30.89	-0.843
supply	cap $\tau^E = 1$	0.01	-46.85	-0.96	26.09	-1.281
supply	transfer	3.39	$+2.72$	14.46	26.56	-0.145

Table 3: Dose-response in the cap intensity, at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$. CEV is the consumption-equivalent variation of the whole scenario relative to the steady state; negative means households would pay that share of permanent consumption to avoid it. The CEV rescaling is $(1+\chi)^{1/4} - 1$ rather than $\chi/4$, since discounted utility scales linearly but the CEV transform does not. Note the transfer's CEV is now negative in every row (see Section 6, caveat on credibility, which this recalibration appears to resolve).

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	-1.300	-2.104	-1.652	-0.144
cap $\tau^E = 0.5$	-0.938	-1.446	-1.254	-0.114
cap $\tau^E = 1$	-0.571	-0.779	-0.851	-0.084
transfer	-0.213	-0.361	-0.450	$+0.172$

Table 4: CEV by discount-factor type (%), at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$. Impatient types hold little wealth and have high MPCs.

5.1 The unit of account is a relabeling, and we verify it

Re-verified at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$: solving the model under both units of account still gives a bit-identical steady state (zero deviation, since $p_H = 1$ at the steady state) and impulse responses (price shock, no policy) agreeing to 2.1×10^{-5} on y and 3.3×10^{-5} on C — about 0.13–0.16% of the respective amplitudes, roughly an order of magnitude larger than the $\sim 0.01\%$ reported at the old calibration, consistent with the larger D^G steady-state derivative feeding more finite-difference noise into the heterogeneous-agent Jacobian at the new ψ_g. **The detailed source decomposition below (the step-size sweep and the grid-refinement check) is carried over from the old calibration and has not been re-run here**; it should be redone before citing the specific benign-noise mechanism at the current calibration, though the order of magnitude (well under 1% of amplitude) is unlikely to change qualitatively.

- For y and C the discrepancy was, at the old calibration, finite-difference noise in the heterogeneous-agent Jacobian. Varying the differentiation step reproduced the characteristic U shape — 4.4×10^{-5} at $h = 10^{-3}$, 1.5×10^{-6} at $h = 10^{-4}$, 5.8×10^{-6} at $h = 10^{-5}$ — truncation on one side, round-off on the other, minimised at the library's default.
- For D^G the discrepancy was invariant to h but halved with grid refinement (5.8×10^{-5} at $n_a = 150$, 3.0×10^{-5} at $n_a = 300$): first-order asset-grid discretisation interacting with a sharp discrete choice.

An earlier implementation left an unexplained residual of about 4% of the IRF amplitude. It was an implementation error, not a property of the model, and it is resolved. Three things must be

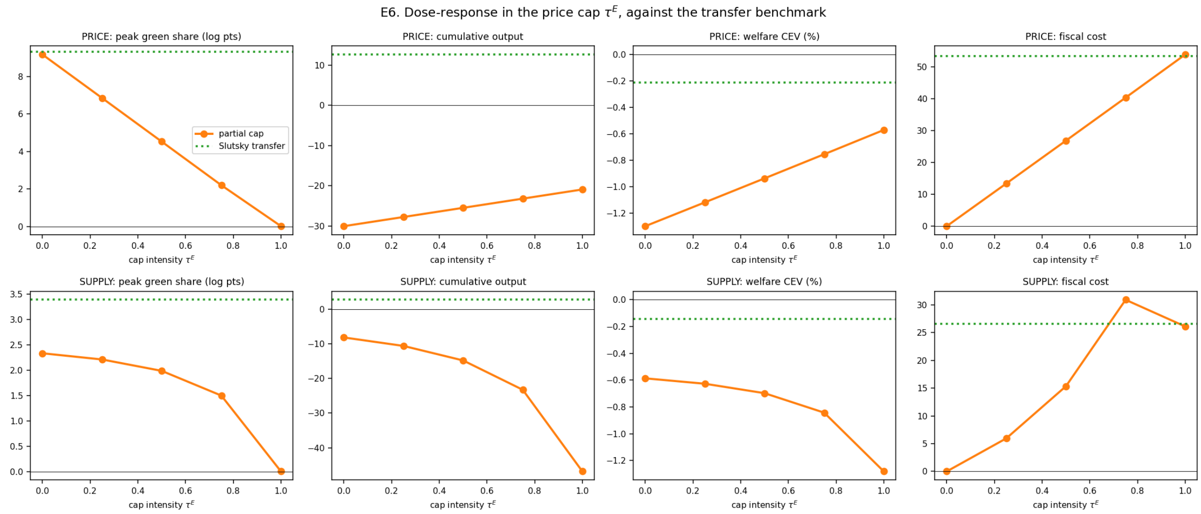

Figure 8: Dose-response in τ^E (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

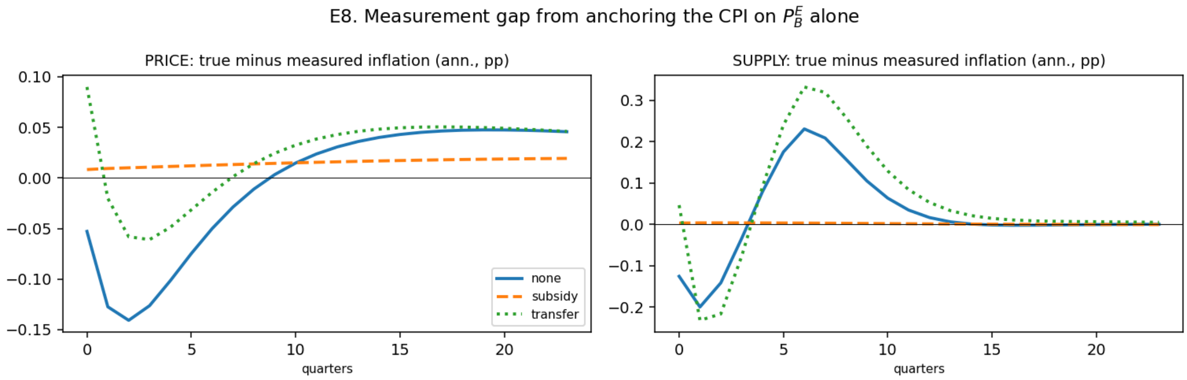

Figure 9: Measurement gap from anchoring the CPI on P_B^E alone: true minus measured annualised inflation, computed ex post outside the DAG (Section 2.5).

converted, and omitting any one of them produces a residual of that order: the CPI-denominated transfers *and* the switching cost, the lagged asset stock in the current account's revaluation term, and *both* legs of every CES demand ratio.

5.2 The adoption margin is strongly non-linear

The first-order solution is still not a good approximation of the adoption channel at the shock sizes used in this literature, though the discrepancy is somewhat smaller with the smoother logit. **The old partial-equilibrium table (household block re-solved non-linearly with price paths frozen at their first-order values, sizes 0.02–1.00) cannot be reproduced here**: the script that produced it (`calibration_moments.py`-adjacent, not part of the current codebase snapshot) no longer exists and needs to be rewritten before that specific comparison can be redone at $\sigma_\varepsilon = 0.05$. What follows instead is the full general-equilibrium nonlinear-vs-linear comparison from `run_linearity_check.py` (Figure 10), which is a strictly harder solve (it lets prices respond) and is therefore the more informative — if more expensive — of the two checks:

Two things follow, both qualitatively unchanged from the old calibration. First, the recession is milder once adoption is solved non-linearly, because a larger share of households escapes the energy bill than the linear solution predicts (D^G's ratio rises with shock size: 1.16, 1.33, 1.67 at

Non-linear / linear, GE	y	C	c_E	D^G	π ann.
price shock, size 0.125	0.963	0.971	0.935	1.159	0.971
price shock, size 0.25	0.914	0.931	0.868	1.330	0.948
price shock, size 0.50	0.781	0.820	0.731	1.667	0.916

Table 5: Ratios at the date of the linear peak, at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$. Inflation and output are damped relative to their linear prediction (ratio < 1) while D^G is amplified (ratio > 1), the same qualitative pattern as at the old calibration. The nonlinear solver does not converge at size 1.00 (the baseline shock); convergence at intermediate sizes between 0.50 and 1.00 has not been tested.

sizes 0.125, 0.25, 0.50). Second, at the baseline shock size (1.00) there is *still no* non-linear GE benchmark, since the solver fails to converge even after raising the Newton iteration cap well above its default — this is not merely a leftover of the old calibration, it persists at $\sigma_\varepsilon = 0.05$. The 5% supply cut used elsewhere in this paper remains motivated by the same concern, though note it is the *price* shock, not the supply shock, whose nonlinear GE solve is being checked here.

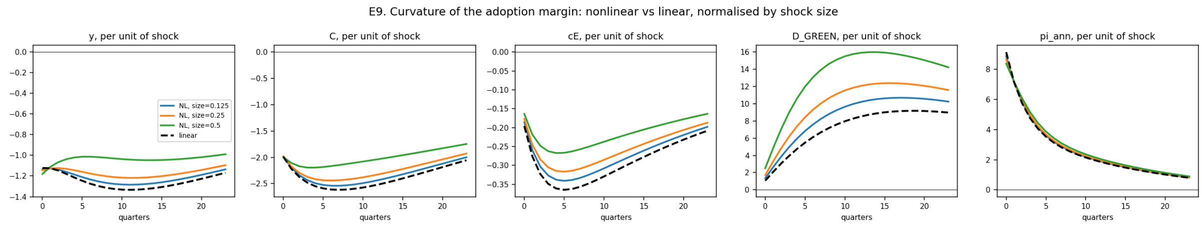

Figure 10: Curvature of the adoption margin: non-linear vs. linear impulse responses, normalised by shock size, for several sizes of the price shock. The linear path (dashed black) coincides with the non-linear one only in the small-shock limit.

5.3 Why the taste scale is doing so much work

It is tempting to read σ_ε as a smoothing device with no economic content. It is not, and the reason is a matter of units.

The derivative of the logit probability is $dP = P\,(d\Delta - d\overline{\Delta})/\sigma_\varepsilon$, so the semi-elasticity of the choice probability with respect to the value gap is exactly $1/\sigma_\varepsilon$, now 20 (was 100). But Δ is a difference of *lifetime* values. With the middle discount factor ($\beta \approx 0.954$), a permanent 1% consumption gain is worth $0.01/(1-\beta) \approx 0.217$ utils, i.e. about 4.3 times σ_ε (was 22 times, at the old $\sigma_\varepsilon = 10^{-2}$). Equivalently, $\sigma_\varepsilon = 0.05$ represents taste heterogeneity worth **0.23% of permanent consumption** (was 0.046%): a five-fold increase in σ_ε is exactly a five-fold increase in this number, since the conversion is linear in σ_ε for fixed β. **The cross-sectional dispersion comparison (the claim that the adoption gap has a standard deviation of 2.63 utils across brown households, $263 \times \sigma_\varepsilon$ at the old calibration) has not been recomputed**: it requires reading the value gap out of the durables stage's internals, which is a nontrivial extraction (see the replication note below on why the naive read is wrong) and was not redone for this pass. Until it is, the "near-deterministic threshold" characterisation should not be repeated verbatim at the new calibration — it may be less extreme now that σ_ε is five times larger, but by how much is not yet known. What *is* unchanged, because it is a mechanical identity independent of σ_ε: $D^{sw}_{ss} = \delta_g D^G_{ss} = 0.25\%$ of households switch each quarter in steady state.

This also explains the curvature, and here the change is unambiguous because it follows from σ_ε alone. The ratio of the second-order to the first-order term in the choice probability is $d\Delta/(2\sigma_\varepsilon)$, so the first-order solution now fails once the shock moves the value gap by $2\sigma_\varepsilon = 0.10$

utils, i.e. about 0.46% of permanent consumption (was 0.09%, a five-fold relaxation). This is consistent with what Table 5 shows directly: the non-linear/linear ratios at a given shock size are closer to 1 than they were at $\sigma_\varepsilon = 10^{-2}$, though still far from 1 at size 0.50. **The specific worked hazard path (0.26% to 28% per quarter along the old size-1.0 non-linear solve) cannot be recomputed**: that solve does not converge at the new calibration either (Section 5 above), so there is no non-linear size-1.0 hazard path to read off at all right now. The curvature still scales as σ_ε^{-2}, so the qualitative mechanism is unchanged; only the point at which it bites has moved.

The two problems are one problem, and this pass is a step, not the answer. Moving σ_ε from 10^{-2} to 0.05 is exactly the direction Section 3.1 recommended, and it visibly relaxes the non-linearity problem (Table 5). But ε_ψ still has not been recomputed at the new value (the identification section's own caveat), so there is no evidence yet that 0.05 is closer to the *identified* value than 10^{-2} was, only that it is less obviously wrong on the payback-period and curvature grounds above. **Calibrating the taste scale against an adoption elasticity remains the highest-return remaining task on this paper**, and until it is done the current value should be presented as a deliberate five-fold correction in the right direction, not as a resolved calibration.

A note for replication. In the solution library, a stage's stored internal value function is the backward *input* to that stage, not its output. The value the discrete choice compares is therefore stored under the *preceding* stage. Reconstructing the choice probabilities from the wrong array produces a switching probability of 0.9997 instead of 0.0024 and looks exactly like a bug in the model.

6 Caveats

We list these because several are load-bearing.

1. **ψ_g and σ_ε are still not separately identified.** σ_ε was moved from 10^{-2} to 0.05 (and ψ_g from 0.253 to 0.538 to hold $D^G_{ss} = 0.05$) on the qualitative grounds of Section 5.3, *not* because ε_ψ was recomputed and matched to an empirical target — that recomputation still requires rewriting `calibration_moments.py`, which is not part of the current codebase. See Sections 3.1 and 5.3. Until an adoption elasticity is imposed, the magnitudes in Table 2 should be read as illustrative, and note that at least one qualitative finding (Result 1's output sign, Result 7's incidence reversal) is now known to be calibration-sensitive, not just its magnitude. The *ordering* of the price-cap-vs-transfer policies (Results 2–6) does not depend on it, since it follows from the relative price alone.
2. **First-order solution.** Section 5. The adoption margin is convex and the first-order solution understates it at the shock sizes previously used (by a factor of about 1.7 in D^G at size 0.50, down from a factor of about 2 at the old calibration), which in turn overstates the recession (nonlinear/linear y ratio 0.78 at size 0.50). This is still the reason the supply cut was reduced to 5%, and the price-shock nonlinear GE solve still fails to converge at the baseline size 1.00 under the new calibration too.
3. **Zero pass-through to the green price.** $\rho_g = 0$ means adopters are *completely* insulated from the fossil shock. This is the upper bound on the adoption channel. A partial pass-through calibrated to observed electricity–gas co-movement would be more defensible; the machinery is in place (ρ_g is a parameter), only the calibration is missing.
4. **The transfer remains very expansionary; its CEV credibility issue has moved, not disappeared.** Output still rises +1.2% (price) on impact, and at $\sigma_\varepsilon = 0.05$ the aggregate transfer CEV is now negative (−0.21%, Table 3) rather than the implausible

+0.13% reported before — but the **patient** type's CEV is still positive (+0.17%, Result 7), i.e. households with little energy exposure and low MPC are still made better off by the crisis-plus-transfer than by the steady state alone. The mechanism is the same as before: full Slutsky compensation combined with debt financing with slow repayment ($\psi_B = 0.04$) and critically **the absence of an investment margin** — Bayer et al. (2026) have capital, which crowds out. The aggregate number is more credible now; the transfer column should still be treated as an upper bound on stabilisation, not an estimate, and the type-level result deserves the same scrutiny the aggregate one used to get.

5. **The supply shock produces a very large price response** (+179% on impact for a 5% quantity cut, up from +93% at the old calibration — the smaller adoption response at $\sigma_\varepsilon = 0.05$ means less substitution away from brown energy, so the world price has to move even further to clear the same 5% quantity cut). With $\eta_E = 0.1$ and energy entering consumption only (prod_E share $= 0$), demand is extremely inelastic, so the price must do all the work. Adding energy to production would damp this considerably.

6. **The CPI is anchored on the brown price.** Section 2.5. The measurement gap now reaches 2.3% of the amplitude of measured inflation (supply shock), smaller than the 6.5% reported at the old calibration because D^G moves less. It does not touch real allocations under the domestic-good numeraire, but it does enter the monetary rule, so the Taylor-rule experiment inherits it.

7. **No within-country energy-share heterogeneity.** Bayer et al. (2026)'s central distributional mechanism is that poor households have higher energy shares. We have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.

8. **Residual steady-state asset market clearing of** 6.9×10^{-7} (recomputed at $\sigma_\varepsilon = 0.05$, essentially unchanged from 6.7×10^{-7} at the old calibration), against 10^{-15} for every other equilibrium condition. Below the tolerance used throughout, but it is the only residual that does not close to machine precision; the likely source is the Jensen gap between exact aggregate energy demand and $\alpha_E C$.

7 Premortem: how Boris would attack this

1. *"Your headline result is mechanical."* The cap fixes P_B^E; adoption depends on P_B^E; therefore adoption dies. This is an identity, not a finding. **Response:** the mechanism is transparent, but the *magnitude* is not — whether the transition loss is second-order or first-order relative to the stabilisation gain is quantitative, and it depends on an adoption elasticity we have not yet identified. The honest framing is that the paper quantifies a trade-off, not that it discovers one.

2. *"Your adoption response is implausibly large."* The green share rises from 5% to 14.2% on the baseline price shock (was 5% to 37% at the old calibration — the smoother logit at $\sigma_\varepsilon = 0.05$ has already made this objection materially weaker). **Response:** the accumulation is internally consistent — the stock identity $D_t^G = (1-\delta_g)D_{t-1}^G + D_t^{sw}$ still holds to machine precision, the brown population never falls below 0.86 (was 0.65), and the switching hazard peaks at 1.33% (was 4.98%), a roughly 5.3-fold rise off its 0.25% steady-state level. We have not recomputed the logit-indifference benchmark (12.6% at the old calibration) that the old hazard peak was compared against. The magnitude is still the stock multiplier $1/\delta_g = 20$ applied to a hazard response, and the *level* of that hazard response is governed by σ_ε, which is still not identified, and the first-order solution still understates it (Table 5). This objection is correct in substance and is weaker in degree than before, but the underlying identification gap (Sections 3.1 and 5) is unresolved, not closed.

3. *"A real cap is not 100%."* **Answered** in Section 4.5: the dose-response is linear under the price shock and convex under the supply shock. The remaining version of the objection is that the French shield capped the *growth rate* of the regulated tariff, not its level, which is a different instrument again.

4. *"Your welfare numbers say the crisis makes people better off."* At $\sigma_\varepsilon = 0.05$ the *aggregate* transfer CEV is negative (-0.21% under the price shock, Table 3), so this specific symptom is gone at the aggregate level. But it has reappeared at the type level: the **patient** type's CEV is positive ($+0.17\%$, Result 7). **Response:** the same fiscal under-discipline that produced the old aggregate result (full Slutsky compensation, slow debt repayment, no investment margin to crowd out) is still present; it now shows up only for the low-MPC, low-energy-exposure group instead of on average. The welfare *ordering* across policies is informative; the type-level and aggregate *levels* both still need the fiscal block disciplined before they are reported as estimates.

5. *"Why is the green price exogenous?"* If everyone adopts, green energy demand rises and its price should rise too. With $\rho_g = 0$ and no green supply curve, adoption has no general-equilibrium cost, which biases the channel upward. This compounds the non-linearity result: the non-linear solution sends adoption higher still, precisely where the missing green supply curve would bite hardest.

6. *"You have no investment and no energy in production."* Both Bayer et al. (2026) and Auclert et al. (2024) have richer supply sides. Ours makes the transfer look too expansionary and the supply shock too inflationary.

7. *"The green durable is not a durable."* It has no stock, no resale, no maintenance — only a switching cost and a breakdown hazard. It is a technology-adoption indicator. Calling it a durable invites comparison to a literature we do not engage.

8. *"The energy-gap block is doing a lot of work."* Booking adopters' saving as a national import saving is one of the two channels through which the adoption margin moves output, the other being the resource cost $\psi_g D^{sw}$ booked as an import (Section 2.6). At the old calibration the saving channel dominated and the margin was expansionary everywhere; at $\sigma_\varepsilon = 0.05$, $\psi_g = 0.538$ the cost channel now dominates under the price shock (Result 1), so "the channel is expansionary" is no longer a model-wide property but a statement that depends on which of the two import terms is larger at the current calibration. The alternative to either choice — a domestic transfer financed by someone — would change the aggregate result materially in either regime. This choice is defended in the text, not in a docstring.

8 Replication

```
ehank/calibration.py    BASE / DURABLE / POLICY layers; make_calibration()
ehank/blocks.py         aggregate blocks, [ARS] vs [NEW] marked; numeraire blocks
ehank/household.py      durable household (StageBlock + LogitChoice + EGM)
ehank/model.py          assembly, steady state, shocks, run()
ehank/deflator.py       share-weighted CPI, computed ex post outside the DAG
ehank/tests.py          non-regression suite (checks 2-5)
ehank/welfare.py        consumption-equivalent variation
run_experiments.py      full matrix + deflator table    -> output/
run_dose_response.py    cap dose-response and welfare   -> output/
run_linearity_check.py  non-linear vs first order       -> output/
```

A run is fully described by its overrides: `run(m, shock_kind='supply', policy='subsidy',` `model_variant='adoption')`. The unit of account is selected once, at `build_model(numeraire=...)`, and the calibration must be built with the same setting. Impulse-response caches are tagged by unit of account, because responses differ across the two by about 0.03% — too little to notice by eye and too much to leave mixed in one table. `test_targets()` is called after every solve because the steady-state solver does not check its own residuals.

Non-regression, run after every change: equilibrium residuals at the steady state and along the transition; the envelope condition $V_a = \partial V/\partial a$; monotonicity of the savings policy, a silent prerequisite of the interpolation routine; the stock-flow identity $D^{sw} = \delta_g D^G$; and the channel-shut nesting test. The envelope converges at a rate of about 1.3, not 2: the asset grid is doubly exponentially spaced, so doubling n_a does not halve the step, and the policy function has a kink at the borrowing constraint.

References

Allcott, H. and M. Greenstone (2012). "Is There an Energy Efficiency Gap?" *Journal of Economic Perspectives* 26(1), 3–28.

Auclert, A., H. Monnery, M. Rognlie, and L. Straub (2024). "Managing an Energy Shock: Fiscal and Monetary Policy." In *Heterogeneity in Macroeconomics: Implications for Monetary Policy*, Series on Central Banking, Analysis, and Economic Policies, Vol. 30, Central Bank of Chile, 39–108.

Bayer, C., A. Kriwoluzky, G. J. Müller, and F. Seyrich (2026). "Redistribution Within and Across Borders: The Fiscal Response to an Energy Shock." *Journal of Political Economy Macroeconomics*, forthcoming. Earlier version: "Hicks in HANK," CEPR DP 18557.

Beresteanu, A. and S. Li (2011). "Gasoline Prices, Government Support, and the Demand for Hybrid Vehicles in the United States." *International Economic Review* 52(1), 161–182.

Berry, S. (1994). "Estimating Discrete-Choice Models of Product Differentiation." *RAND Journal of Economics* 25(2), 242–262.

Busse, M. R., C. R. Knittel, and F. Zettelmeyer (2013). "Are Consumers Myopic? Evidence from New and Used Car Purchases." *American Economic Review* 103(1), 220–256.

Grigolon, L., M. Reynaert, and F. Verboven (2018). "Consumer Valuation of Fuel Costs and Tax Policy: Evidence from the European Car Market." *American Economic Journal: Economic Policy* 10(3), 193–225.

Hausman, J. A. (1979). "Individual Discount Rates and the Purchase and Utilization of Energy-Using Durables." *Bell Journal of Economics* 10(1), 33–54.

Iskhakov, F., T. H. Jørgensen, J. Rust, and B. Schjerning (2017). "The Endogenous Grid Method for Discrete-Continuous Dynamic Choice Models with (or without) Taste Shocks." *Quantitative Economics* 8(2), 317–365.

Langot, F., S. Malmberg, F. Tripier, and J.-O. Hairault (2026). "The Macroeconomic and Redistributive Effects of the French Tariff Shield." *Journal of Political Economy Macroeconomics*, forthcoming. Earlier version: CEPREMAP Working Paper 2305.

Muehlegger, E. and D. S. Rapson (2022). "Subsidizing Low- and Middle-Income Adoption of Electric Vehicles: Quasi-Experimental Evidence from California." *Journal of Public Economics* 216, 104752.